

Ritvik Gupta

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AI/ML Engineer with 2+ years of experience delivering production grade ML and LLM solutions, including forecasting, anomaly detection, and large scale text summarization systems. Expert in Python, SQL, and GCP, with hands on experience building scalable data pipelines, prompt driven LLM applications, and automated workflows. Demonstrated impact in improving decision making through enhanced data quality, optimized model performance, and efficient, cost aware system design.

Education

Boston University | *Master of Science, Applied Data Analytics*

2023 - 2025

SRM Institute of Science and Technology

2019 - 2023

Bachelor of Technology (B.Tech.), Computer Science and Engineering

Skills

- **Machine Learning & LLMs:** LLM Applications, Prompt Engineering, Forecasting, Anomaly Detection, XGBoost
- **Programming:** Python, SQL, PySpark
- **Data Engineering:** ETL Pipelines, BigQuery, Data Validation
- **Cloud & Tools:** GCP, AWS (S3, EMR), MLflow, Docker, Kafka, Union AI

Work Experience

Prestanda Consulting | *AI ML Engineer*

Jan 2026 - Present

- Built a production scale LLM powered review summarization system using Gemini Flash & Pro, generating context aware summaries from large scale review data via structured prompts.
- Designed a cron based, trigger driven regeneration framework with custom threshold logic to update summaries only when new reviews meaningfully impact content, minimizing redundant LLM inference.
- Developed scalable data pipelines on GCP (BigQuery) orchestrated via Union AI, enabling reliable processing and near real time summary updates.
- Improved system efficiency by reducing unnecessary recomputation and optimizing summary refresh cycles for better cost and latency.

The Chicago Education Advocacy Cooperative | *Data Science Fellow*

Jun 2025 - Dec 2025

- Built and validated forecasting models used for budgeting and planning, improving forecast accuracy by 25% across financial and operational data.
- Implemented anomaly detection and data validation pipelines that increased precision by 15% and enabled earlier identification of data and operational issues.

Boston University | *Graduate Teaching Assistant*

Jan 2024 - May 2025

- Taught 80+ students in Python, SQL, regression modeling, hypothesis testing, and feature engineering using real world retail and operations datasets.
- Designed hands on labs focused on forecasting, EDA, and data storytelling, significantly improving applied analytics and model interpretability skills.

ENGAGEathon | *Data Analyst Intern*

May - Aug 2024

- Designed KPI frameworks and Power BI dashboards for inventory flow, supplier performance, and SLA monitoring, contributing to an 18% reduction in SLA misses.
- Automated ETL pipelines across 9+ systems, standardizing feature generation and improving reliability of operational reporting used by business teams.

Technical Projects

Scalable Recommendation Engine with PySpark & AWS

- Developed feature pipelines and trained hybrid recommender models (ALS, FAISS, K-Means) on 10M+ customer-product interaction records for large scale personalization.
- Ran distributed batch training and scoring on AWS EMR with S3 based storage, optimizing Spark execution and cutting recommendation pipeline latency by ~35%.

Unstructured Data Extraction & Ranking Pipeline (Python + NLP)

- Implemented LLM powered NLP feature extraction pipelines to clean noisy text, normalize tokens, and convert unstructured content into structured, ML ready feature vectors, with MLflow used to track feature experiments and downstream model performance.
- Developed a scoring and ranking framework using LLM derived features with anomaly validation, leveraging MLflow based experiment tracking to compare ranking strategies and automatically surface high value, reliable business signals.

Publications

- Analyzing Sector Performance & Investment Strategies: A Decade of FTSE Data — IJEFM, 2025
- Face Mask Detection Using Deep Learning CNN Architecture — IJSER, 2022
- Payment System for University — IJSER, 2023